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RESEARCH ARTICLE

Real-Time Human Group Detection and Clustering in Crowded Environments Using Enhanced Multi-Object Tracking

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ABSTRACT Group detection is a critical yet challenging task in video-based applications such as surveillance analysis, especially in crowded and dynamic environments where complex pedestrian interactions occur. Traditional trajectory-based methods often struggle with occlusions and overlapping behaviors, leading to inaccurate group identification. To address these limitations, we propose a novel algorithm that integrates an optimized YOLOv8 model with DeepSORT tracking, enhancing both detection accuracy and real time performance. Our approach uniquely combines high-precision object detection with stable multi-object tracking, ensuring consistent identification of individuals and groups over time, even in high-density scenarios. Additionally, we introduce an innovative method of constructing an adjacency matrix by integrating Euclidean distances and bounding box diagonal ratios, which is transformed into a graph to intricately analyze and predict complex group dynamics in real time. Experimental results on real-world airport CCTV footage demonstrate that our method significantly outperforms existing approaches, achieving higher precision and recall rates. Furthermore, the algorithm operates efficiently on standard hardware, indicating strong practical feasibility for real-time applications in public spaces. While challenges such as misclassification due to incomplete data annotations and occlusions remain, our study showcases the potential of integrating spatial and temporal data to advance real-time group detection and tracking, aiming to improve crowd management systems in public spaces.

INDEX TERMS Multi-object tracking, visual surveillance, group detection.

I. INTRODUCTION

In recent years, group detection and clustering have emerged as critical areas of research in video surveillance and crowd analysis, driven by the increasing need for public safety and efficient crowd management [1]. A group is typically defined as a collection of individuals who exhibit coordinated behavior or share common traits that bind them into a cohesive unit. As technology advances, identifying groups within crowded environments and analyzing their behavioral patterns have become increasingly essential for understanding pedestrian dynamics and managing crowds effectively. However, challenges arise in densely populated settings, where individual behaviors become difficult to distinguish, leading to inaccurate group identification. Traditional group detection methods have primarily relied on trajectory-based approaches [2], [3], which track individual movement paths over time to identify group patterns. While effective in

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less crowded environments, these methods face limitations in complex and high-density settings with occlusions and overlapping behaviors. Advancements in deep learning have introduced powerful techniques for group detection [4], [5], [6], [7], excelling at capturing intricate patterns and subtle variations in pedestrian interactions. However, these models often analyze behavior based on individual video frames, limiting their ability to account for the temporal continuity of pedestrian actions and potentially missing important temporal relationships that influence group dynamics. To address these issues, we propose a novel approach that integrates YOLOv8 [8] and DeepSORT [9], [10], two of the most widely recognized models in real-time tracking. YOLOv8 provides high-precision object detection, accurately identifying pedestrians even in high-density environments. DeepSORT offers robust multi-object tracking capabilities, ensuring consistent identification of individuals across video frames. By combining these two methods, our approach captures both the spatial and temporal aspects of pedestrian movement, overcoming issues caused by occlusions and overlapping behaviors. A key innovation of our method is the construction of an adjacency matrix using Euclidean distances and bounding box diagonal ratios between pedestrians. This adjacency matrix is transformed into a graph structure that models the spatial relationships that form group dynamics. By integrating spatial proximity and movement patterns, our method enables accurate real-time group identification and clustering. This approach effectively leverages advanced detection accuracy and temporal continuity, addressing challenges that existing trajectory-based and deep learning methods face in maintaining consistent identification in crowded and dynamic scenarios. The contributions of this paper are summarized as follows:

- We propose a novel method for real-time group detection and tracking in crowded environments by integrating YOLOv8 and DeepSORT, combining high-precision detection with stable multi-object tracking.
- We develop an innovative approach using Euclidean distances and bounding box diagonal ratios to construct an adjacency matrix, which is then transformed into a graph to analyze and predict complex group dynamics in real-time.

II. RELATED WORK

A. TRAJECTORY-BASED GROUP DETECTION

Trajectory-based approaches [11] have played a pivotal role in analyzing pedestrian behavior and understanding group formation. These methods typically focus on the physical and social interactions among individuals, predicting group formation based on their movement patterns [7], [12]. For example, frameworks have been developed that leverage multivariate stochastic differential equations to model individual acceleration in continuous time, taking into account both the positions and velocities of surrounding entities. In crowded scenes, methods such as Anchor-based Manifold Ranking (AMR) [13] have been introduced to analyze topological relationships among individuals. AMR employs a coherent merging strategy to identify groups, even in highly populated environments [14], [15], which enhances the ability to detect dynamic group interactions.

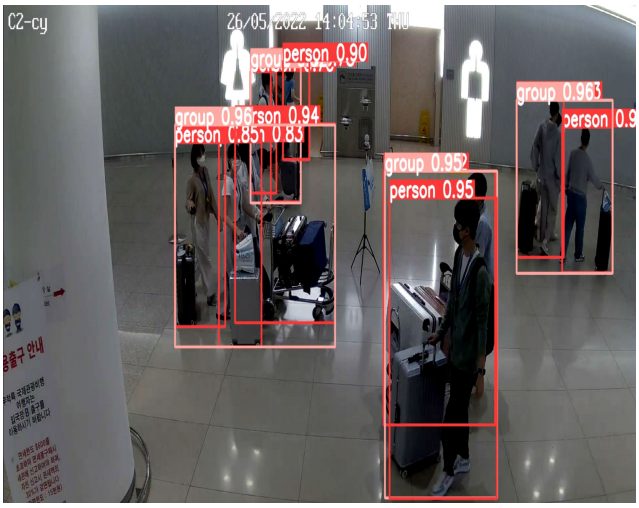


FIGURE 1. Visualization of person and group detections using the custom-trained YOLOv8 model. Each bounding box represents either a person or a group class with corresponding confidence scores, applied in a crowded airport environment.

B. CHALLENGES AND ADVANCES IN TRAJECTORY-BASED METHODS

While trajectory-based methods offer valuable insights, particularly through pose estimation and interaction modeling [11], they face considerable challenges in dense, dynamic environments. The presence of occlusions and complex social behaviors can diminish the effectiveness of these approaches. Recent advancements have sought to address these issues by integrating human body pose trajectories with novel detection algorithms. For example, a self-supervised framework [16] leverages multi-head networks with social relation feature embedding and self-attention mechanisms to improve group detection. These developments highlight the potential of trajectory-based methods but also emphasize the need to handle noise and estimation errors effectively.

To further enhance robustness, researchers have incorporated Long Short-Term Memory (LSTM) [17] and graph transformer techniques [18]. These methods encode pedestrian movement as node features and geometric distances as edge features, enabling more accurate and real-time group analysis [19]. Although trajectory-based approaches continue to be integral in understanding group dynamics, future work must address their limitations in highly populated and complex environments to ensure scalability and precision.

C. DEEP LEARNING-BASED APPROACHES FOR GROUP DETECTION

The use of deep learning in group detection has grown significantly, driven by the need to cluster pedestrians in dense

environments for applications such as crowd flow monitoring and behavior analysis [20], [21]. With advancements in neural networks, techniques such as Generative Adversarial Networks (GANs) [4], Deep Neural Networks (DNNs) [5], and Graph Convolutional Networks (GCNs) [6] have been applied to detect interaction groups with impressive accuracy. GANs, for instance, have been used to unsupervisedly detect social groups, while DNNs integrate pedestrian features for group detection. Furthermore, ResNet-X and Graph Neural Networks (GNNs) have explored high-dimensional feature aggregation and spatial relationships between groups [22].

Despite these advancements, deep learning-based approaches come with inherent challenges, such as the need for substantial training data and high computational costs. Additionally, many current methods rely heavily on static Euclidean distances, which limits their effectiveness in dynamic, real-world scenarios. To overcome these challenges, we propose an enhanced method that models pedestrians as nodes within a graph structure, incorporating both spatial and temporal factors. This enables real-time analysis and prediction of group dynamics, improving the scalability and flexibility of deep learning-based approaches.

III. PROPOSED METHOD

A. BACKGROUND

This research proposes an approach that integrates the YOLOv8 and DeepSORT algorithms to effectively detect and track pedestrians and groups in complex and dynamic environments. Traditional detection and tracking systems struggle to maintain accuracy in densely populated or overlapping situations. Therefore, this research explores leveraging the strengths of YOLOv8 and DeepSORT to enable consistent detection and grouping, even in these challenging conditions. The proposed algorithm is structured into three primary steps: i) Pedestrian Detection, ii) Group Relationship Initialization via Multi-Object Tracking, and iii) Group Interaction Analysis and Final Group Identification. This workflow is illustrated in Figure 2, and the subsequent subsections provide a detailed explanation of each step.

B. PEDESTRIAN DETECTION

In the first step, YOLOv8 [8] is used to detect pedestrians and groups within an airport dataset. YOLOv8 is chosen for its high accuracy and fast processing, making it effective in complex environments. The group coordinates in the dataset are converted into YOLO format to ensure consistency. This transformation is performed using the following equation:

$$\left(\hat{x}_i, \hat{y}_i, \hat{w}_i, \hat{h}_i\right) = \left(\frac{x_i}{W}, \frac{y_i}{H}, \frac{w_i}{W}, \frac{h_i}{H}\right), \tag{1}$$

where W and H represent the image width and height, (x_i, y_i) denotes the original group center coordinates, and w_i and h_i represent the group's width and height, respectively. After detection, the YOLOv8 model provides bounding boxes for both pedestrians and groups. The detected pedestrian and group bounding boxes provide the foundational data for the multi-object tracking phase.

C. GROUP RELATIONSHIP INITIALIZATION VIA MULTI-OBJECT TRACKING

The second step involves tracking the detected objects using the DeepSORT algorithm [10]. The main challenge in this phase is maintaining consistent tracking of overlapping or occluded objects in dynamic environments. To address this, a bounding box re-detection algorithm was integrated into DeepSORT to preserve tracking consistency even when IoU-based matching fails. The following IoU [23] formula is used to match pedestrian and group bounding boxes:

$$\mathrm{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}, \tag{2}$$

where $|B_p \cap B_g|$ represents the intersection area between the pedestrian bounding box B_p and the group bounding box B_g, and $|B_p \cup B_g|$ denotes the union area of the two bounding boxes. If the IoU exceeds a threshold θ, the pedestrian is matched with the group. This IoU-based matching is essential for initializing group relationships and ensuring consistency in matching.

Algorithm 1 Direct IoU Calculation for Group Matching

Require: B_p: Array of pedestrian bounding boxes, B_g: Array of group bounding boxes, θ: IoU threshold
Ensure: P: List of grouped pedestrian pairs

```
M ← calculate_iou(B_p, B_g)
I ← argmax(M, axis = 1)
V ← max(M, axis = 1)
Initialize G as an empty dictionary
for i = 0 to length(B_p) − 1 do
    if V[i] > θ then
        G[ID of B_g[I[i]]].append(B_p[i])
    end if
end for
Initialize P as an empty list
for each id ∈ G do
    for each b ∈ G[id] do
        P.append((b, id))
    end for
end for

return P
```

Bounding boxes with an IoU value exceeding a certain threshold θ are prioritized for matching, thereby establishing spatial relationships between pedestrians and groups. Algorithm 1 outlines this IoU-based matching process. In this algorithm, IoU scores are calculated between each pedestrian bounding box B_{p_i} and group bounding box B_{g_j}. If the IoU score exceeds the threshold θ, the corresponding pedestrian-group pair is matched. The algorithm adds these matched pairs to the M_i list, enabling more accurate

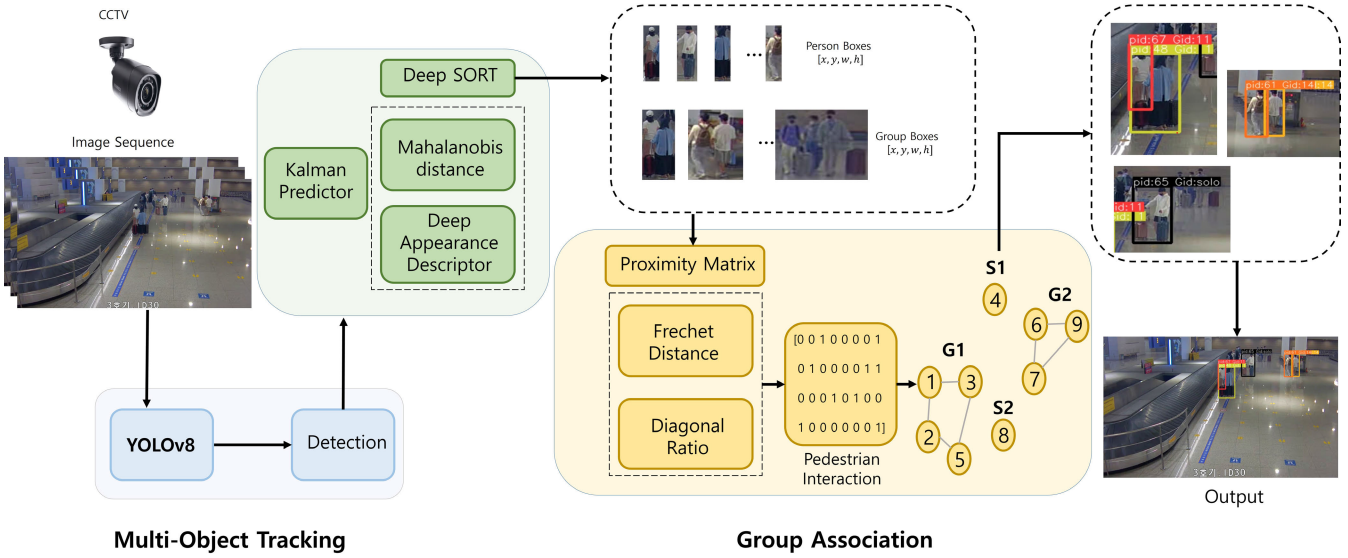


FIGURE 2. The process of associating groups using the DeepSORT architecture. The YOLOv8 model is used for detecting individuals and groups, followed by DeepSORT, which tracks these entities based on Kalman prediction, Mahalanobis distance, and deep appearance descriptors.

analysis of relationships between pedestrians and groups and facilitating the grouping process.

Using the results from M_i, we generate the final grouped results for pedestrians sharing the same group ID. To do this, we first initialize an empty dictionary called G_{ID} to store the pedestrians associated with each group ID. For each pedestrian-group pair, the pedestrian's ID and the group's ID are stored in G_{ID}. Finally, to output the pedestrians sharing the same group ID, we initialize a list called P, where the final grouped results are stored.

Based on the matching values P, derived from Algorithm 1, we conduct a detailed analysis to more precisely delineate group relationships among pedestrians. Specifically, we evaluate whether individuals within each group are genuinely interacting by examining their trajectories and spatial distributions. This approach integrates the physical proximity and behavioral patterns of pedestrians to more clearly define the final group configurations.

D. GROUP INTERACTION ANALYSIS AND FINAL GROUP IDENTIFICATION

In the final step of our methodology, we refine group assignments by analyzing interactions between pedestrians within each group. This process is particularly challenging in crowded environments due to the dynamic nature of pedestrian movements. To enhance the precision of our interaction analysis, we employ two key metrics: the diagonal ratio and Fréchet distance, combined with Euclidean distances to form a comprehensive adjacency matrix. This approach allows us to capture real-time dynamic interactions and effectively addresses the limitations of traditional group detection techniques.

A fundamental advancement in our approach is the construction of an adjacency matrix using Euclidean distances and bounding box diagonal ratios, which is then transformed into a graph structure. This methodology allows us to

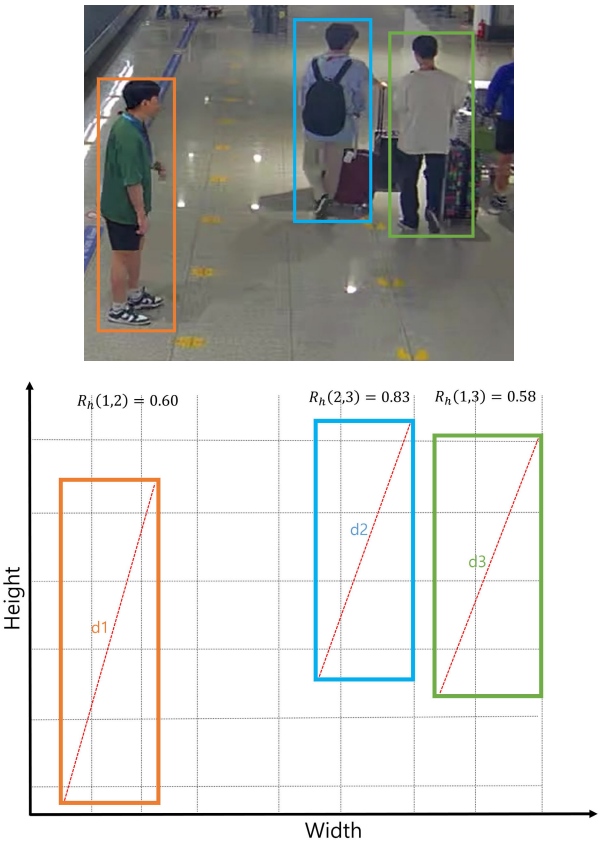


FIGURE 3. The visualization demonstrates the body size differences among three pedestrians through diagonal ratios. On the left, diagonal lengths are determined by the pedestrians' widths and heights, with notable differences in their ratios: 0.60 between the first two and 0.83 between the last two, indicating varying extents of size disparity.

capture real time dynamic interactions between pedestrians, effectively overcoming limitations of traditional techniques in crowded environments.

1) DIAGONAL RATIO

The diagonal ratio is introduced to reduce distortions caused by perspective or body orientation when comparing object sizes. This metric provides a stable size comparison by comparing the bounding box diagonals. The diagonal ratio $R_{i,j}$ is defined as:

$$R_{i,j} = \frac{\min(d_i, d_j)}{\max(d_i, d_j)} \tag{3}$$

where d_i and d_j represent the diagonal lengths of pedestrians i and j, respectively. Unlike horizontal or vertical ratios, the diagonal ratio minimizes distortions caused by perspective and body orientation, enabling consistent size comparison across frames.

To further refine interaction analysis, we calculate the diagonal angle between bounding boxes to assess pedestrian orientation. This angle, denoted $T_{i,j}$, is calculated using the arccosine of the angle between diagonal vectors:

$$T_{i,j} = \cos^{-1}\left(\frac{\vec{d}_i \cdot \vec{d}_j}{|\vec{d}_i||\vec{d}_j|}\right) \tag{4}$$

The combined metrics of size and orientation, denoted as $I_{i,j}$, allow us to better understand interaction dynamics. Analyzing relative size and orientation differences provides insight into how pedestrians are positioned and interacting within a group. Figure 3 visually illustrates body size differences among three pedestrians using diagonal ratios, showcasing the practical application of this metric in assessing size disparities.

2) Fréchet DISTANCE

Another critical measurement in group identification is recognizing sustained interaction patterns between pedestrians. To capture temporal and spatial movement patterns, we use the discrete Fréchet distance [24] between pedestrian trajectories:

$$\delta_{dF}(P, Q) = \max d(p_{a_i}, q_{c_i}) \tag{5}$$

where p_{a_i} and q_{c_i} are pairs of sampled points on trajectories P and Q at the same time, with a_i and c_i being their indices. The distance function $d(p_{a_i}, q_{c_i})$ is defined as the Euclidean distance between each pair of sampling points:

The Fréchet distance, grounded in Euclidean calculations, efficiently represents physical proximity and movement consistency over time, distinguishing fleeting interactions from sustained group movement. By integrating Fréchet distance, we achieve an accurate identification of social groups based on interactions beyond individual frames.

3) GROUP CLUSTERING

Once the diagonal ratio and Fréchet distance are calculated, we use these values to construct an adjacency matrix D:

$$D_{i,j} = I_{i,j} \oplus F_{p,q} \tag{6}$$

where $I_{i,j}$ contains pedestrian size and orientation information, and $F_{p,q}$ includes Fréchet distance data. By applying a Depth-First Search (DFS) algorithm [25]to this matrix, pedestrians with high similarity are clustered together, and those within the same cluster are assigned the same group ID, completing the final group identification process.

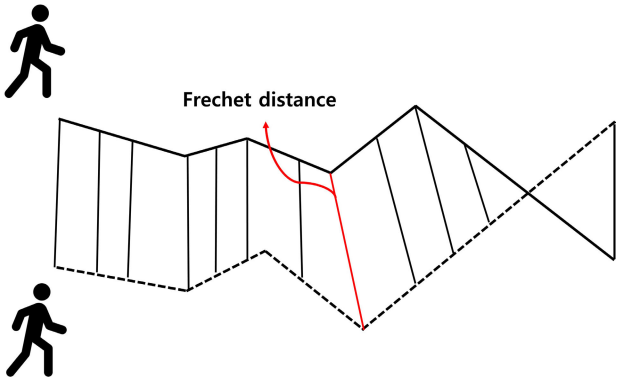


FIGURE 4. Discrete Fréchet distance.

IV. EXPERIMENTS

In this section, we present a thorough evaluation of the proposed group detection and clustering algorithm. The experiments are conducted using a real-world dataset, and performance is measured using various metrics. We also discuss implementation details, comparisons with different configurations, and visualizations of the results.

A. DATASET

To evaluate the performance of our algorithm, we used a video dataset captured in a real airport environment. The dataset consists of video footage from various locations within the airport, collected via fixed CCTV cameras. The dataset includes 90,490 images, split into 78,647 for training, 8,743 for validation, and 1,100 for testing. Each video frame is annotated with group IDs, individual IDs within each group, and bounding box information for each person. This dataset offers a diverse set of challenges, such as varying lighting conditions, crowd densities, and dynamic interactions, providing a robust benchmark for assessing group detection and tracking in realistic scenarios. By using this dataset, we comprehensively evaluate the algorithm's ability to handle complex real-world conditions.

B. COMPARISON WITH EXISTING DATASET

The multimodal JRDB dataset consists of 64 minutes of data collected from various indoor and outdoor environments of a university campus by the JackRabbot robot. This dataset

TABLE 1. Results of metrics for the compared methods at IoU = 0.2.

Video	IoU Matching			IoU Matching + Group Association		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
1	0.76	0.73	0.74	**0.85**	**0.75**	**0.80**
2	**0.78**	0.69	0.72	0.77	**0.75**	**0.76**
3	0.77	0.67	0.72	**0.90**	**0.76**	**0.84**
4	0.78	**0.72**	0.75	**0.82**	0.69	**0.76**
5	0.86	**0.84**	**0.85**	**0.88**	0.68	0.77
6	0.79	0.68	0.74	**0.84**	**0.70**	**0.76**

TABLE 2. Results of metrics for the compared methods at IoU = 0.3.

Video	IoU Matching			IoU Matching + Group Association		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
1	0.76	0.77	0.76	0.77	**0.78**	**0.77**
2	**0.77**	0.70	0.72	0.75	**0.70**	0.72
3	0.74	0.62	0.68	**0.85**	**0.78**	**0.81**
4	0.65	**0.58**	0.61	**0.66**	0.57	**0.62**
5	0.70	0.63	0.66	**0.75**	**0.66**	**0.70**
6	0.62	0.54	0.57	**0.70**	**0.65**	**0.67**

TABLE 3. Results of metrics for the compared methods at IoU = 0.4.

Video	IoU Matching			IoU Matching + Group Association		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
1	0.44	0.42	0.43	**0.56**	**0.44**	**0.49**
2	0.35	0.30	0.32	**0.62**	**0.34**	0.44
3	0.43	**0.33**	**0.38**	**0.56**	0.28	0.37
4	0.47	0.25	0.33	**0.55**	**0.34**	**0.42**
5	**0.63**	**0.51**	**0.56**	0.51	0.33	0.40
6	**0.57**	0.49	**0.53**	**0.53**	**0.51**	0.52

TABLE 4. Comparison of inference times for different grouping methods.

Video	Inference Time (ms)	
	IoU Matching	IoU Matching + Group Association
1	36.3	39.4
2	37.1	37.2
3	37.3	38.5
4	36.5	38.8
5	36.8	38.5
6	36.6	37.1
Avg	**36.77**	**38.25**

includes a wide range of human postures and interactions. It provides over 2.4 million 2D bounding boxes from panoramic 360-degree images captured by five stereo RGB cameras, and over 1.8 million 3D bounding boxes from point-clouds captured by two 16-array LiDAR sensors, all of which are interconnected. However, the JRDB lacks sufficient information for analyzing social human activities, leading to the development of the JRDB-Act dataset [26], which includes additional annotations for individual actions and social groups. This new dataset is the only one capable of performing a variety of human-related tasks and has been meticulously reviewed by professional annotators to ensure the consistency and quality of the annotations.

C. EVALUATION METRICS

To evaluate performance, the IoU metric is used to measure how closely detected group bounding boxes match the actual group bounding boxes. IoU calculates the ratio of the intersection area to the union area of the two bounding boxes. In this study, the IoU threshold is set at 0.8, meaning that any detections exceeding this value are considered true positives (TP). This threshold was determined with reference

to Table 5. Based on this setup, the methods for calculating Precision, Recall, and the F1-Score are explained below.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

Precision, calculated as the ratio of true positives (TP) to the sum of true positives and false positives (FP), indicates the algorithm's capability to correctly identify actual groups from all groups it detects. A higher precision score reflects the algorithm's efficiency in minimizing false positives.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

Recall, defined as the ratio of true positives to the sum of true positives and false negatives (FN), measures the algorithm's ability to capture all relevant groups present in the data. This metric helps assess how well the algorithm minimizes missed detections, or false negatives, in the detection process.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

The F1-score, calculated as the harmonic mean of precision and recall, provides a balanced measure of the algorithm's accuracy by considering both its precision and recall values. This metric serves as a comprehensive indicator of the model's performance in reducing errors while maintaining detection completeness. These metrics collectively allow us to evaluate the proposed approach's effectiveness across different environmental conditions, ensuring its practical applicability and robustness in accurately detecting social groups.

D. IMPLEMENTATION DETAILS

We implemented the proposed approach using a consistent set of parameters across all video datasets. Object detection was performed using YOLOv8 with 100 training epochs and a learning rate of 0.01. In the tracking phase, the parameters were set as follows: detection threshold = 0.3, maximum object age = 50, minimum hits = 1, and IoU threshold = 0.1 for frame associations. The grouping algorithm was integrated into the detection and tracking pipeline to ensure real-time group detection throughout the video sequences. The system was run on a single NVIDIA 3090 Ti GPU, with each video taking approximately eight hours to process. These settings were consistently applied to ensure uniformity across datasets and evaluation consistency.

E. PERFORMANCE COMPARISONS

1) PERFORMANCE EVALUATION BASED ON IoU THRESHOLDS

To evaluate the performance of the proposed group detection method, an analysis was conducted on the matching between groups and individual pedestrians using the matching threshold values of 0.2, 0.3, and 0.4 from Eq 2 for persons and groups. This analysis was performed using basic IoU matching as well as IoU matching with an added group association feature, and the results can be found in Tables 1, 2 and 3. In Table 1, when the group association feature was added, improvements in Precision, Recall, and F1-Score were observed in most videos, with particularly significant performance enhancement in Video 3. In Table 2,

TABLE 5. In Table 1, the evaluation for Videos 1, 2, and 3 under the "IoU Matching + Group Association" method is detailed according to different IoU thresholds, assuming the highest values are at IoU = 0.8.

	Threshold	Precision	Recall	F1-Score
Video 1	0.7	0.82	0.70	0.75
	0.8	**0.85**	**0.75**	**0.80**
	0.90	0.83	0.73	0.77
Video 2	0.7	0.75	0.73	0.74
	0.8	**0.77**	**0.75**	**0.76**
	0.90	0.75	0.72	0.73
Video 3	0.7	0.88	0.74	0.80
	0.8	**0.90**	**0.76**	**0.84**
	0.90	0.88	0.74	0.80

TABLE 6. Experiment with the "IoU Matching + Group Association" method at IoU = 0.2.

Video	Precision	Recall	F1-Score
Video 1	0.34	0.31	0.32
Video 2	0.46	0.44	0.42
Video 3	0.39	0.35	0.37

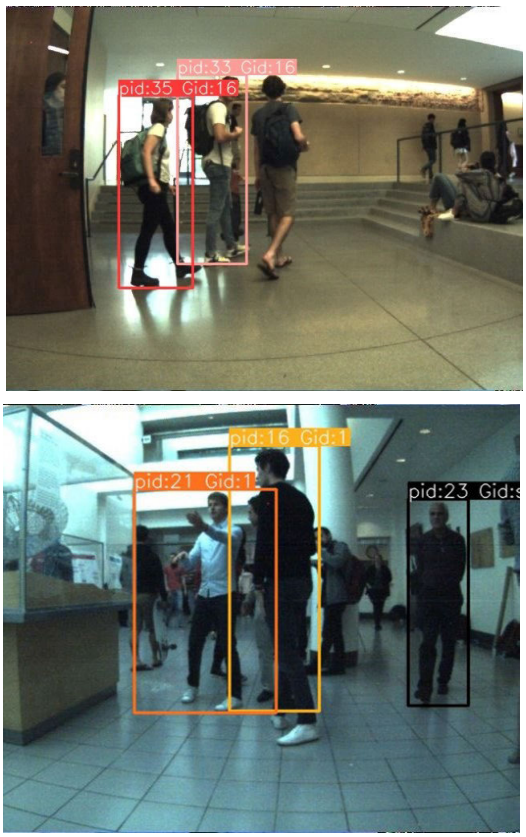


FIGURE 5. Experiment results using the JRDB dataset.

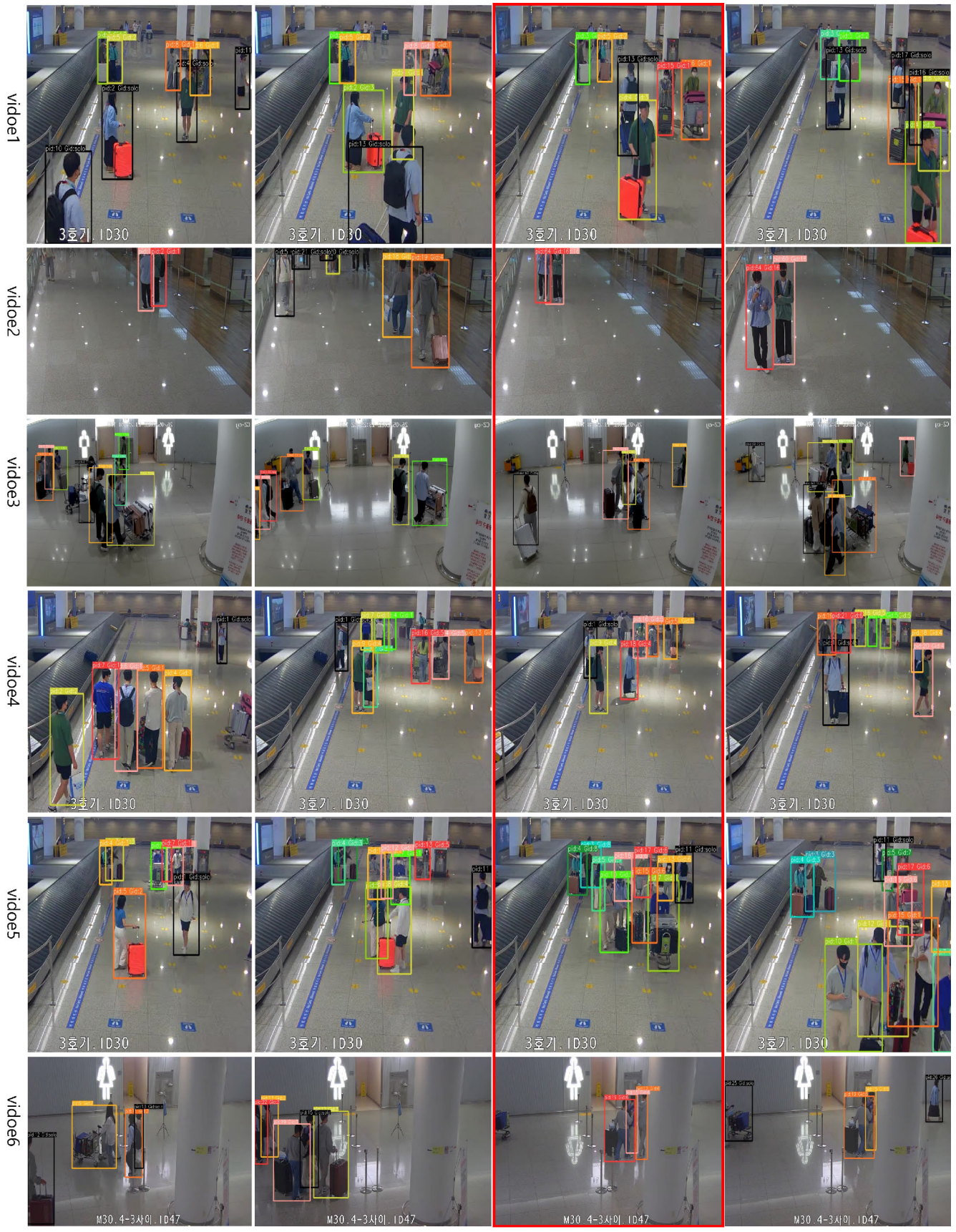


FIGURE 6. Qualitative evaluation.

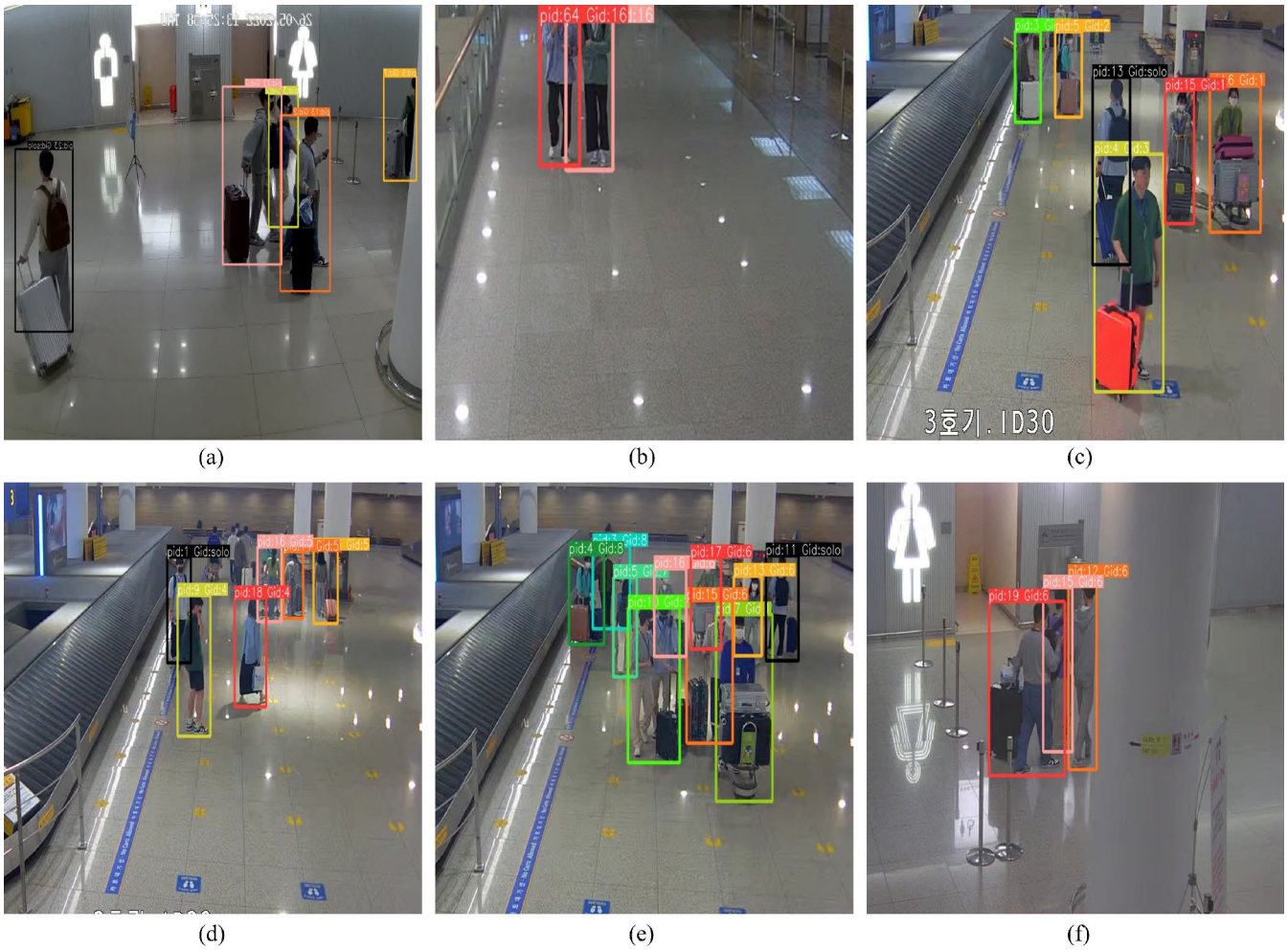


FIGURE 7. The detail qualitative evaluation.

performance improvements were also noted, but the consistency of the F1-Score was somewhat lower. This is because the higher threshold setting imposes stricter matching criteria on the algorithm, which may lead to failing to recognize actually similar groups. However, this setting can contribute to increased accuracy in grouping in scenes with complex interactions. On the other hand, at the highest Table 3, all performance metrics were generally lower. This is due to the very strict matching criteria underestimating the actual similarities between groups, leading to misclassification of entities with similar behavioral patterns. Such results can hinder the flexibility of the analysis and prevent the useful group association feature from effectively demonstrating its potential. Overall, the experimental results with varying IoU thresholds show that while the group association feature generally improves grouping performance, setting the threshold too high can reduce the algorithm's effectiveness.

2) COMPARATIVE EXPERIMENTS UNDER DIFFERENT FRAME RATES

According to the analysis of Table 4 in the paper, there was no significant difference in inference times between *IoU Matching* and *IoU Matching + Group Association* methods. This indicates that the added Group Association feature had a minimal impact on the algorithm's processing speed. Consequently, the Group Association function can serve as a useful tool that enables more detailed analysis without compromising inference efficiency.

F. VISUALIZATION

In the experiments conducted across six diverse video scenarios shot in an airport, our group recognition method has effectively identified and visually represented complex multi-activity groups in Figure 6. This figure visualizes one-quarter of the total frames for each video. Particularly, a specific frame highlighted in red in Figure 6 is enlarged and detailed in Figure 7. Each scenario incorporates various social situations such as restroom entrances and immigration checks, with groups displayed using color-coded bounding boxes. Each bounding box indicates group membership, with 'pid' and 'gid' labels specifying the IDs of individuals and groups, respectively. Notably, our method showcases its capability to swiftly adjust group members when group interactions temporarily change. For example, as seen in



(a) Errors caused by occlusions



(b) Errors due to solo individuals detected as groups

FIGURE 8. **Some cases of failing to identify groups.**

Video 5, when people in proximity are momentarily considered a group and then disperse, our system reassigns them to their original groups. This ability proves the excellence of our approach in effectively capturing and responding to social interactions.

G. ABLATION STUDY

1) METRIC COMPARISON

Evaluation metrics are measured based on whether the predicted bounding boxes exceed a specific IoU threshold with the group bounding boxes. Experimental results have shown that the performance peaks at an IoU threshold of 0.8, which can be verified in Table 5. We also analyzed results at IoU thresholds of 0.7 and 0.85. At an IoU of 0.7, the performance was slightly lower than at 0.8, and at 0.90, there was a further decline in performance. This suggests that the precision required at higher IoU thresholds may be more challenging to achieve in real-world applications. Such analysis provides critical information for determining the optimal IoU threshold necessary to optimize the performance and accuracy of group detection algorithms.

2) DATASET COMPARISION

The JRDB dataset is the only publicly available dataset that provides group bounding box information, offering invaluable resources for object recognition and tracking research in complex video scenarios. To align this dataset with the model and evaluation metrics I proposed, I converted its label format to the YOLO format. The model was designed using parameters such as diagonal ratio and Fréchet distance to correct visual distortions that may occur from various CCTV angles. This design is especially crucial for effective object recognition and tracking in large-scale surveillance environments, such as airports.

However, the JRDB dataset predominantly offers frontal views, which significantly differ from the CCTV perspectives for which our model was specifically designed. This discrepancy hindered the full utilization of the dataset's visual characteristics, leading to lower performance in the experiments based on my predefined evaluation metrics compared to previous results obtained from airport datasets. These results are thoroughly detailed in Table 6, indicating that the model, optimized for specific surveillance environments, was less effective under conditions such as the frontal views provided by the JRDB dataset. This outcome underscores the need for further adjustments and enhancements to the model to ensure broader applicability across various viewing angles and environments. Additionally, these differences and results can also be seen in Figure 5

H. DISCUSSION

In this work, we introduce a novel lightweight deep learning framework that effectively combines YOLOv8 and DeepSORT to enhance group detection capabilities in a variety of environmental settings. Our methodology has proven to be effective, yet, throughout our research, we have encountered significant challenges that hinder optimal performance, as illustrated in Figure 8. One major issue arises from the dependency on the YOLO detection framework, which due to inadequately annotated data, can mistakenly classify individuals as incorrect group members. Such misclassifications weaken the algorithm's robustness and significantly impact the accuracy of group detection. As seen in equation(7) and equation(8), precision and recall are calculated based on true positives (TP), false positives (FP), and false negatives (FN). This means that an increase in FP due to misdetections reduces precision, while an increase in FN decreases recall, negatively affecting the overall performance assessment. Additionally, occlusions caused by overlapping groups, such as those labeled "gid5" and "pid1," can lead to non-detection of some group members. In complex environments, these issues are exacerbated, presenting considerable challenges to achieving precise and consistent group identification. To address these challenges, it is imperative to integrate both spatial context and temporal dynamics within the detection process. Furthermore, the augmentation of our datasets with comprehensive annotations detailing group behaviors is essential for improving detection accuracy. In light of the limitations observed, future research will focus on developing more sophisticated methods of group recognition that seamlessly integrate spatial and

temporal data, aiming to refine the accuracy and reliability of our group detection processes.

V. CONCLUSION

In this paper, we presented a novel algorithm designed to improve group detection accuracy in video surveillance, particularly in crowded environments where traditional methods face limitations due to complex pedestrian interactions. Our approach utilizes temporal relationships between video frames to better capture the subtle dynamics of group behavior in real time. By integrating a custom YOLOv8 model with DeepSORT, we achieved robust tracking and reliable identification of both individual pedestrians and groups. A key innovation of our method is the construction of an adjacency matrix using Euclidean distances and bounding box ratios, which is transformed into a graph structure for comprehensive, real-time group dynamics analysis. Tested on complex airport CCTV footage, this approach outperforms existing methods, particularly in environments with diverse pedestrian behaviors and unpredictability. By leveraging YOLOv8's precision alongside DeepSORT's tracking stability, our method facilitates reliable detection and tracking, addressing challenges like occlusion and overlapping behaviors. While occasional misclassifications can occur due to incomplete annotations, the YOLOv8-DeepSORT combination proves advantageous in crowded settings where maintaining object stability is crucial. Future enhancements will incorporate additional contextual data and dynamic adjustments to further refine group identification.

DATA AVAILABILITY

All data used in this study have been obtained with the appropriate permissions and do not pose any copyright issues.

REFERENCES

[1] J. C. Turner, "Towards a cognitive redefinition of the social group," Res. Colloq. Social Identity Eur. Lab. Social Psychol., Université de Haute Bretagne, Rennes, France, Tech. Rep., 1978.
[2] T. Li, H. Chang, M. Wang, B. Ni, R. Hong, and S. Yan, "Crowded scene analysis: A survey," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 3, pp. 367–386, Mar. 2015.
[3] H. Swathi, G. Shivakumar, and H. S. Mohana, "Crowd behavior analysis: A survey," in *Proc. Int. Conf. Recent Adv. Electron. Commun. Technol. (ICRAECT)*, Mar. 2017, pp. 169–178.
[4] K. Tan, L. Xu, Y. Liu, and B. Luo, "Small group detection in crowds using interaction information," *IEICE Trans. Inf. Syst.*, vol. 100, no. 7, pp. 1542–1545, 2017.
[5] H. C. Fernando and P. De Leyn, "Surgical management of patients considered marginally resectable," in *IASLC Thoracic Oncology*. Amsterdam, The Netherlands: Elsevier, 2018, pp. 314–317.
[6] R. Akbari, H. Pardilla, and Z. Zulfah, "In junior high school boys, a comparison of hollow sprint and acceleration sprint methods on the sprinter 100 meters," *Inspiree, Indonesian Sport Innov. Rev.*, vol. 3, no. 1, pp. 47–55, Dec. 2021.
[7] J. Sun, J. Yao, Y. Liu, S. Lin, Z. Xu, and L. Li, "Formation of hollow CoNi-S nanowedges arrays via sulfidation-etch of ZIF-L for advanced hybrid supercapacitor," *ChemistrySelect*, vol. 5, no. 2, pp. 660–667, Jan. 2020.
[8] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
[9] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," 2017, *arXiv:1703.07402*.
[10] Y. Du, Z. Zhao, Y. Song, Y. Zhao, F. Su, T. Gong, and H. Meng, "StrongSORT: Make DeepSORT great again," *IEEE Trans. Multimedia*, vol. 25, pp. 8725–8737, 2023.
[11] S. K. Pang, J. Li, and S. J. Godsill, "Detection and tracking of coordinated groups," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 47, no. 1, pp. 472–502, Jan. 2011.
[12] F. Setti, C. Russell, C. Bassetti, and M. Cristani, "F-formation detection: Individuating free-standing conversational groups in images," *PLoS ONE*, vol. 10, no. 5, May 2015, Art. no. e0123783.
[13] M. Chen, Q. Wang, and X. Li, "Anchor-based group detection in crowd scenes," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2017, pp. 1378–1382.
[14] J. Shao, C. C. Loy, and X. Wang, "Learning scene-independent group descriptors for crowd understanding," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 27, no. 6, pp. 1290–1303, Jun. 2017.
[15] M. S. Zitouni, A. Sluzek, and H. Bhaskar, "Visual analysis of socio-cognitive crowd behaviors for surveillance: A survey and categorization of trends and methods," *Eng. Appl. Artif. Intell.*, vol. 82, pp. 294–312, Jun. 2019.
[16] J. Li, R. Han, H. Yan, Z. Qian, W. Feng, and S. Wang, "Self-supervised social relation representation for human group detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 142–159.
[17] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 961–971.
[18] R. Han, H. Yan, J. Li, S. Wang, W. Feng, and S. Wang, "Panoramic human activity recognition," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 244–261.
[19] S. Jahangard, M. Hayat, and H. Rezatofighi, "Real-time trajectory-based social group detection," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2023, pp. 1901–1908.
[20] C. Shao, X. Wang, X. Wang, C. Du, C. Dang, and S. Liu, "Cooperative dispatch of wind generation and electric vehicles with battery storage capacity constraints in SCUC," *IEEE Trans. Smart Grid*, vol. 5, no. 5, pp. 2219–2226, Sep. 2014.
[21] J. Shao, N. Dong, and Q. Zhao, "A real-time algorithm for small group detection in medium density crowds," *Pattern Recognit. Image Anal.*, vol. 28, no. 2, pp. 282–287, Apr. 2018.
[22] N. Robinson, B. Tidd, D. Campbell, D. Kulić, and P. Corke, "Robotic vision for human–robot interaction and collaboration: A survey and systematic review," *ACM Trans. Hum.-Robot Interact.*, vol. 12, no. 1, pp. 1–66, Mar. 2023.
[23] G. I. O. Union, "A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Sep. 2019, pp. 658–666.
[24] I. Kang, H. Choi, Y. J. Yoon, J. Park, S.-S. Kwon, and C. Park, "Fréchet distance-based cluster analysis for multi-dimensional functional data," *Statist. Comput.*, vol. 33, no. 4, p. 75, Aug. 2023.
[25] R. Tarjan, "Depth-first search and linear graph algorithms," *SIAM J. Comput.*, vol. 1, no. 2, pp. 146–160, Jun. 1972.
[26] M. Ehsanpour, F. Saleh, S. Savarese, I. Reid, and H. Rezatofighi, "JRDB-act: A large-scale dataset for spatio-temporal action, social group and activity detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 20951–20960.



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